# Marketing Analytics

## Generative AI, Consumer Choice, and Regulation

By Nathan Yang

Session 14

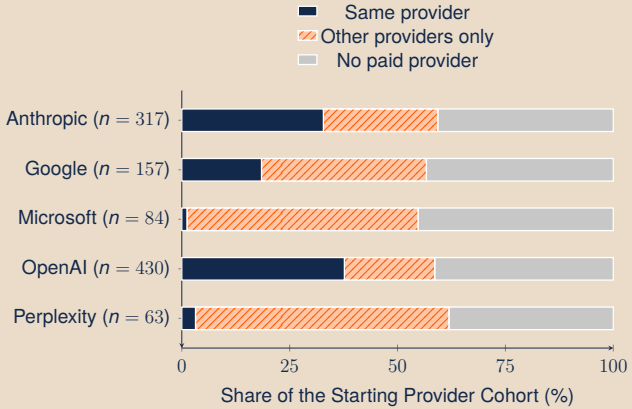
# A Subscription-Growth Decision

- A firm wants to increase paid renewals.
- The evidence includes subscription histories, interfaces, and customer outcomes.
- A higher renewal rate can arise through improved value or greater cancellation friction.

# Public Attention and Paid Demand

- Pageviews measure attention to selected pages.
- Paid subscriptions measure a specific payment relationship.
- User, provider, and category denominators differ.

# A Fixed Starting Cohort

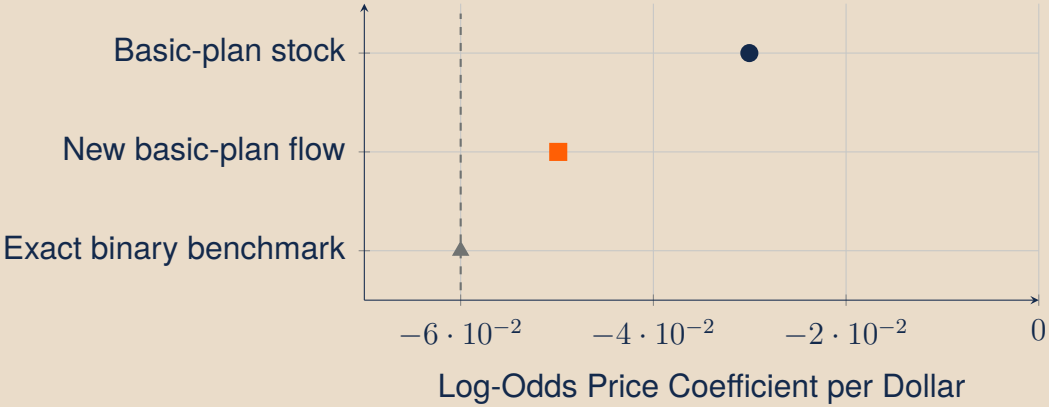



Entirely synthetic subscription panel. Provider labels identify simulated options.

# Subscriber Stocks and New Flows

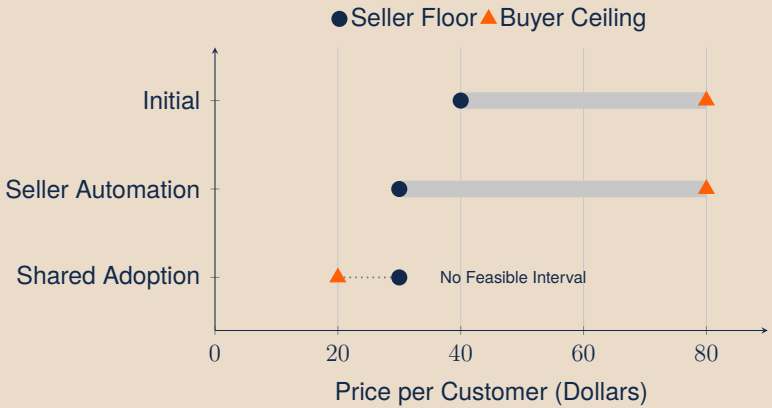
- A subscriber stock includes past acquisition and survival.
- A new-subscription flow can include first purchases and returns.
- A price coefficient depends on the outcome and behavioral model.

# A Known Model Is a Useful Verification Case




Synthetic demand exercise with a known binary-model benchmark.

# Seller Costs and Customer Alternatives

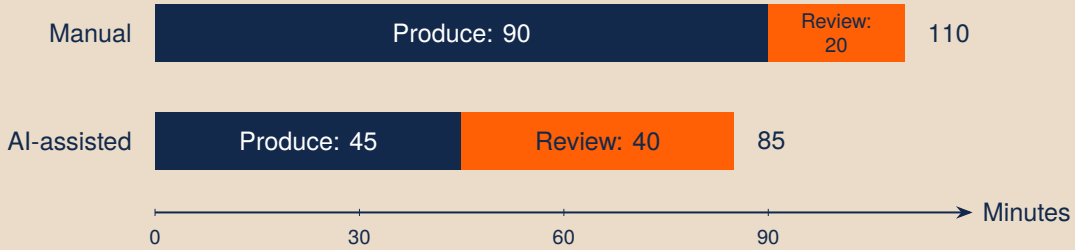



Hypothetical dollars per customer, conditional on 50 sales.

# AI Workflow Quality

- Measure the completed analysis and its validity.
- Include generation, review, and repair time.
- Use comparable tasks and independent quality criteria.

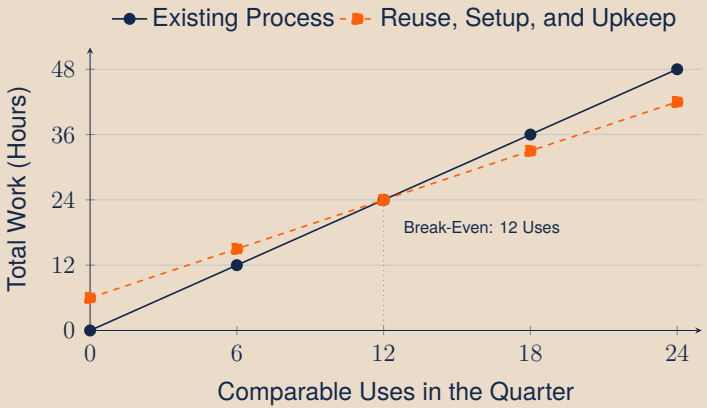
# Review Time Is Part of Production




Drafting time falls 50%; total time falls 22.7%.
Task and completed-work quality held fixed.

Synthetic workflows meet the same stated quality requirement.

# Reuse Has a Break-Even Workload




Hypothetical equal-quality work: six setup hours, with adaptation included.

# Model Updates and Unresolved Tasks

- Connect an update to a recorded difficulty and a feasible comparison.
- Evaluate the frozen challenger and current procedure on fresh cases.
- Include setup, correction, and maintenance in the adoption decision.

# A Choice Procedure Can Change Renewal

- A direct offer and an offer preceded by questions can produce different choices.
- The customer still makes the final selection.
- An independent value measure determines whether an additional renewal benefits that customer.

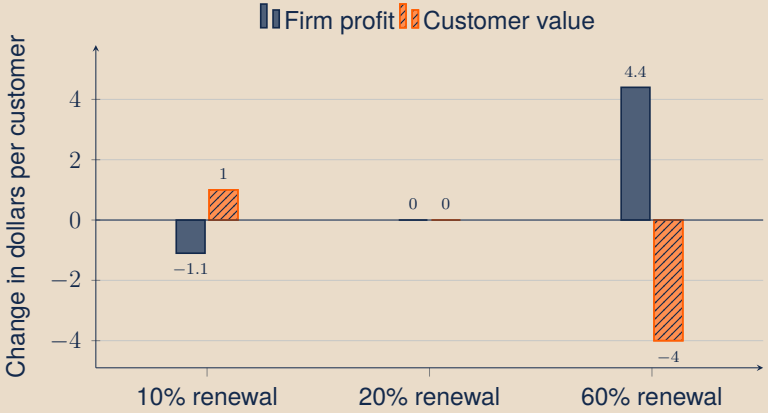
# Opposed Interests

Suppose renewal yields firm margin $\nu > 0$ and costs the customer $\sigma > 0$ relative to declining.

$$\Delta\Pi = \nu(s - s_0), \qquad \Delta V = -\sigma(s - s_0)$$

A higher renewal probability raises profit and lowers customer value under these stated stakes.

# Profit and Customer Value Can Move Oppositely




Synthetic probabilities and fixed stakes. Direct renewal is 20%.

# Experimentation Selects the Commercial Winner

- A fixed set of procedures has stable renewal probabilities.
- Independent randomized samples estimate each probability.
- With opposed interests, accurately selecting the largest rate also selects the largest loss in that set.

# A Cancellation Evaluation

- Include a direct-cancellation comparison.
- Record whether optional retention content can be refused before exposure.
- Measure completion, elapsed time, subsequent charges, and customer value.

# Rule Status and Scope

- A statute, binding rule, interpretive guide, and proposal have different authority.
- Record jurisdiction, covered conduct, exceptions, and effective date.
- Use the current primary source before turning a result into a business recommendation.

# Pricing, Reviews, and Delivery

- Price-display rules require attention to the covered industry.
- Reviews presented as customer experiences require truthful provenance.
- Housing-ad delivery can create discrimination concerns beyond targeting choices.

# Pricing Language and Regulatory Evidence

- Shelf infrastructure, price rules, and consumer effects are separate objects.
- The FTC’s January 2025 report describes research on pricing intermediaries.
- Specify the practice and applicable authority before making a legal claim.

# A Welfare Comparison

Synthetic stakes distinguish profit from customer value.

```
direct, selected = 0.20, 0.60
firm_margin, customer_loss = 11, 10
firm_gain = firm_margin * (selected-direct)
value_change = -customer_loss * (selected-direct)
assert round(firm_gain, 2) == 4.40
assert round(value_change, 2) == -4.00
```

Higher combined surplus can coexist with a customer loss under these assumptions.

# Coding Exercise: A Marketing Evaluation

- Calculate reuse break-even and the conditional price interval.
- Compute profit and customer value for alternative retention procedures.
- Propose an interface change and the evidence needed to evaluate it.

# A Decision With Consumer Outcomes

- State the commercial target and customer outcome.
- Identify the procedure the customer can accept, refuse, or revise.
- Connect the proposed action to current rules and supporting evidence.